

PART 14 OF 18

Enterprise AI Governance

Operating Model, RACI Matrices, AI Platform Teams, PromptOps, AgentOps, Risk Committees, Responsible AI Boards, and Policy Lifecycle Management

ENTERPRISE AI CONTROL ARCHITECTURE

Implementation Guide for Production AI Systems • 2026

14.1 The Enterprise AI Operating Model

Deploying autonomous AI agents at enterprise scale requires a governance operating model that does not exist in most organizations today. Traditional IT governance assumes deterministic software; AI governance must contend with systems that make judgments, exhibit behavioral variability, learn from experience, and may develop emergent behaviours. The governance model must create clear accountability without requiring every decision to be made by a committee.

Governance Design Principle: Speed and safety are not opposites. Well-designed governance enables faster, higher-confidence deployment by providing clear frameworks for decisions that would otherwise require case-by-case escalation. The goal is governance that empowers teams to move quickly within safe boundaries, not governance that creates bureaucratic checkpoints.

14.2 Core AI Governance Roles

14.2.1 AI Platform Team

The AI Platform Team owns the shared infrastructure that all agent deployments depend on: the agent orchestration platform, the identity and authorization systems, the observability infrastructure, the tool registry, and the memory governance systems. The team is the internal equivalent of a cloud provider—product teams consume capabilities; the platform team ensures they are secure, reliable, and compliant.

AI Platform Team Responsibilities

- Design and operate the agent execution platform (orchestration, sandboxing, runtime controls)
- Maintain the Identity Broker, Capability Broker, and AI Policy Engine
- Operate the Tool Registry and tool approval workflow
- Own the observability infrastructure (telemetry pipeline, behavioral analytics, anomaly detection)
- Define and maintain platform-level security standards and hardening guides
- Provide approved agent templates and starter kits to development teams

14.2.2 AI Security Team (AISecOps)

The AI Security Team is the security engineering and operations function specialized in AI-specific threats. Distinct from the general information security team, AISecOps requires expertise in: LLM security, prompt injection, agent behavioral analysis, AI supply chain security, and AI compliance frameworks.

- Lead threat modeling for all AI agent deployments
- Conduct pre-deployment security reviews and red-teaming
- Own the AI SOC function (ADR monitoring, incident response, forensics)
- Maintain MITRE ATLAS threat intelligence for the enterprise
- Perform quarterly AI security assessments and capability audits
- Own AI security policies and standards (AI security baseline, tool security standards)

14.2.3 PromptOps Team

PromptOps manages the lifecycle of system prompts, prompt templates, and constitutional constraints used in production agents. Prompts are first-class configuration artifacts with security implications—a poorly constructed system prompt is a security vulnerability.

- Develop and maintain approved prompt template library with security-reviewed baseline prompts
- Own prompt versioning, testing, and deployment pipeline (prompt CI/CD)
- Conduct prompt injection resistance testing for all new prompts
- Monitor prompt effectiveness and safety metrics in production
- Manage prompt signing and cryptographic verification infrastructure
- Provide prompt security training to development teams

14.2.4 AgentOps Team

AgentOps handles the operational lifecycle of deployed agents: deployment, monitoring, performance optimization, capacity planning, and retirement. AgentOps is the agent equivalent of DevOps—responsible for ensuring agents run reliably at scale.

14.2.5 ModelOps Team

ModelOps manages the lifecycle of AI models used in enterprise deployments: model evaluation, fine-tuning oversight, model versioning, model retirement, and performance monitoring. ModelOps is responsible for ensuring that the models underlying enterprise agents meet quality and safety standards.

14.2.6 MemoryOps Team

MemoryOps manages enterprise AI memory systems: schema design, access control configuration, retention policy implementation, integrity monitoring, and memory incident response. MemoryOps treats AI memory as a regulated data store equivalent to a database containing sensitive enterprise data.

14.3 Governance Committees and Boards

14.3.1 AI Risk Committee

The AI Risk Committee has authority to approve high-risk agent deployments, set enterprise-wide AI risk tolerance thresholds, and make risk-acceptance decisions for AI capabilities that have no precedent. Membership: CISO, CTO, Chief Risk Officer, Chief Privacy Officer, General Counsel, Business Unit Leaders. Meets monthly; emergency sessions for critical incidents.

14.3.2 Responsible AI Board

The Responsible AI Board provides ethical oversight of AI deployments: ensuring agents treat users fairly, respect privacy, avoid discriminatory outcomes, and align with corporate values. The board reviews any agent deployment affecting customers, employees, or third parties at significant scale.

14.4 RACI Matrix: Agent Deployment Lifecycle

Activity	AI Platform	AI Security	AgentOps	PromptOps	Risk Committee	Business Owner
Platform architecture decisions	A	C	I	I	I	I
Security threat modeling	C	A	I	C	I	I
System prompt design	C	C	I	A	I	R
Tool approval	C	A	C	I	I	I
Agent deployment approval (low-risk)	A	C	R	C	I	I
Agent deployment approval (high-risk)	C	R	R	C	A	R
Production monitoring	C	C	A	I	I	I
Security incident response	C	A	R	C	I	I
Agent retirement	C	C	A	C	I	R
Compliance reporting	R	R	R	R	A	I
Memory policy management	C	C	A	I	I	I
Risk committee escalation	I	R	I	I	A	R

RACI Key: R = Responsible, A = Accountable, C = Consulted, I = Informed

14.5 AI Policy Lifecycle Management

14.5.1 Policy as Code for AI Governance

AI governance policies must be encoded in machine-readable policy languages (Cedar, Rego) that can be version-controlled, tested, peer-reviewed, and deployed through a CI/CD pipeline. Policy-as-code enables: audit trail of policy changes, automated testing before policy deployment, rollback capability if a policy causes unintended consequences, and continuous compliance verification.

Stage	Activities
Draft	Policy authored in Cedar/Rego; documented intent; initial review by policy author
Peer Review	AI Security + Legal + Privacy review; automated policy linting and conflict detection
Simulation Testing	Policy evaluated against synthetic agent behavior dataset; impact analysis

Stage	Activities
Staged Rollout	Policy deployed to 5% of agents; monitoring for unexpected impacts for 48 hours
Full Rollout	Policy deployed to all agents; continuous monitoring for compliance gaps
Review	Quarterly review against new threat intelligence, regulatory changes, and operational feedback
Deprecation	Policy marked deprecated; migration path provided; 90-day sunset period